

Q1. Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30,000	60
C2	60,000	40
C3	50,000	70
C4	28,000	65
C5	65,000	35

Question: Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability.

Comparative Analysis & Solution:

- **Performance on Dataset:** The data segments into two natural groups: low-income/high-spending (C1, C4) and high-income/lower-spending (C2, C5, with C3 being moderate-high). K-Means partitions data into k spherical clusters by minimizing within-cluster sum of squares around centroids. Hierarchical (Agglomerative) clustering builds a bottom-up dendrogram without requiring a pre-specified k, merging C1 and C4 first due to minimal Euclidean distance.
- **Computational Complexity & Scalability:** K-Means has a time complexity of $O(n \cdot k \cdot I \cdot d)$ and linear memory $O(n)$, scaling efficiently to large enterprise customer bases. Hierarchical clustering requires computing an $n \times n$ distance matrix with time complexity $O(n^2 \log n)$ or $O(n^3)$ and memory $O(n^2)$, making it computationally prohibitive for large datasets but highly interpretable for small samples ($n=5$).

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.20	0.50	0.30
D2	0.10	0.60	0.30
D3	0.80	0.10	0.10
D4	0.75	0.15	0.10

D5

0.20

0.40

0.40

Question: Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.

Comparative Analysis & Solution:

- **Hard vs. Soft Clustering:** D3 and D4 form a clear cluster dominated by Word1. Documents D1, D2, and D5 share high proportions of Word2 and Word3, with D5 (0.2, 0.4, 0.4) residing directly along the decision boundary between topic distributions. K-Means performs hard assignment, strictly forcing D5 into a single cluster and losing boundary nuance.
- **Handling Overlapping Topic Profiles:** Expectation-Maximization (EM with Gaussian Mixture Models) performs soft clustering by estimating posterior probabilities $P(\text{Cluster}_k \mid \text{Document}_i)$. For overlapping document D5, EM assigns fractional membership (e.g., 55% Cluster A, 45% Cluster B) and models elliptical/covariance shapes, making EM vastly superior for document collections exhibiting mixed themes.

Q3. Dataset: Geographic Data

Point	Latitude (°N)	Longitude (°E)
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question: Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results.

Comparative Analysis & Solution:

- **Distance Formulations:** Euclidean (L2) distance $d = \sqrt{(\Delta\text{Lat})^2 + (\Delta\text{Long})^2}$ calculates straight-line geometric distance. Manhattan (L1) distance $d = |\Delta\text{Lat}| + |\Delta\text{Long}|$ computes city-block distance along orthogonal axes.
- **Impact on Cluster Formation:** Points form two spatial clusters: Northern (P1, P2, P4) and Southern (P3, P5). Euclidean distance assumes isotropic spatial dispersion, creating circular boundary partitions (optimal for aerial/direct line-of-sight analysis). Manhattan distance reflects grid-constrained transit networks and is less sensitive to extreme coordinate deviations in any single axis.

Q4. Dataset: Retail Products

Product	Sales (₹)	Purchase Frequency
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P1	50,000	20
P2	30,000	10
P3	52,000	22
P4	25,000	8
P5	32,000	12

Question: Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset.

Comparative Analysis & Solution:

- **Methodological Difference:** Agglomerative (Bottom-Up) initializes each of the 5 products as singleton clusters and merges the nearest pairs iteratively (e.g., merging high-value tier P1 & P3 first, then low-tier P2 & P5). Divisive (Top-Down) begins with all 5 products in one master cluster and recursively splits based on maximal inter-cluster dissimilarity (e.g., separating {P1, P3} from {P2, P4, P5}).
- **Suitability for Retail Dataset:** Agglomerative is much more suitable for this small product catalog (n=5) due to lower algorithmic overhead $O(n^2)$ compared to divisive approaches, which require $2^{(n-1)}-1$ partition evaluations at each split step unless constrained by heuristic flat-clustering algorithms like K-Means.

Q5. Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question: Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types.

Comparative Analysis & Solution:

- **Handling Mixed Data Types (Numerical + Categorical):** The dataset contains continuous numerical features (Price, Rating) and nominal categorical features (Category A/B). Centroid-based methods (e.g., standard K-Means) compute

arithmetic means, which are mathematically undefined for categorical labels unless adapted into K-Prototypes or combined with One-Hot encoding.

- **Effectiveness & Clustering Paradigm:** Density-based techniques (e.g., DBSCAN) cluster points based on local point density (ϵ -neighborhood and MinPts) rather than mean vectors. By utilizing custom generalized distance functions like Gower's Distance or HEOM (Heterogeneous Euclidean-Overlap Metric), density-based clustering can directly process mixed data and identify arbitrary cluster shapes while isolating noisy outliers.